Obtener la transformada de Laplace:

1. $\mathscr{L}\{te^{-3t}cos(3t)\}$
2. $\mathscr{L}\{te^{2t}cos(6t)\}$
3. $\mathscr{L}\{t^3e^t)\}$
4. $\mathscr{L}\{t^2sinh(t)\}$
5. Revisar, estudiar y concluir acerca del ejemplo 8 (aplicación de un voltaje periódico) del zill (página 300 del documento, página 343 del pdf.)

Usar la transformada de Laplace para resolver los siguientes problemas de valor inicial.

1. $y'' + y = f(t)$
$\begin{cases} y(0) = 1 \\ y'(0) = 0 \end{cases}$
$f(t) = \begin{cases} cos(4t) & si \quad 0 \leq t < \pi \\ 0 & si \quad t \geq \pi \end{cases}$
2. $y'' + 16y = f(t)$
$\begin{cases} y(0) = 0 \\ y'(0) = 1 \end{cases}$
$f(t) = \begin{cases} 1 & si \quad 0 \leq t < \dfrac{\pi}{2} \\ sin(t) & si \quad t \geq \dfrac{\pi}{2} \end{cases}$
3. $y'' + 9y = cos(3t)$
$\begin{cases} y(0) = 2 \\ y'(0) = 5 \end{cases}$
4. $y' - y = te^tsin(t)$
$\{ y(0) = 0$
5. $2y'' + ty' - 2y = 10$
$\begin{cases} y(0) = 0 \\ y'(0) = 0 \end{cases}$

Usar transformada de integrales:

1. $\mathscr{L}\{1 * t^3\}$
2. $\mathscr{L}\{e^{2t} * sin(t)\}$
3. $\mathscr{L}\left\{t\int_0^t \tau e^\tau d\tau\right\}$
4. $\mathscr{L}\left\{\int_0^t sin(\tau)cos(t-\tau)d\tau\right\}$

Determinar la transformada inversa de Laplace:

1. $\mathscr{L}^{-1}\left\{\dfrac{e^{-2s}}{s^3}\right\}$
2. $\mathscr{L}^{-1}\left\{\dfrac{(1+e^{-2s})^2}{s+2}\right\}$
3. $\mathscr{L}^{-1}\left\{\dfrac{e^{-2s}}{s^2(s-1)}\right\}$
4. $\mathscr{L}^{-1}\left\{\dfrac{e^{-s}}{s(s+1)}\right\}$
5. $\mathscr{L}^{-1}\left\{\dfrac{1}{(s+2)^3}\right\}$
6. $\mathscr{L}^{-1}\left\{\dfrac{1}{s^2+2s+5}\right\}$
7. $\mathscr{L}^{-1}\left\{\dfrac{2s+5}{s^2+6s+34}\right\}$
8. $\mathscr{L}^{-1}\left\{\dfrac{(s+1)^2}{(s+2)^4}\right\}$